A Multi-Method Geophysical, Hydrogeological and Hydrological Investigation of Post-Mining Environmental Effects at the Abandoned Lead Mine, Pretoria, South Africa

Abstract

The abandoned Edendale Lead Mine (ELM), situated on the eastern outskirts of Pretoria within the Silverton Formation of the Pretoria Group, represents a significant legacy of South Africa's 6100 derelict and ownerless mine sites, having been decommissioned in 1974 prior to the introduction of modern environmental legislation. This study employed a multi-method approach integrating magnetic, gravity, electrical resistivity tomography (ERT), and seismic refraction tomography (SRT) geophysical surveys, together with hydrogeochemical characterisation of mine water and surface water, geochemical speciation modelling using PHREEQC, and salt tracer dilution flow measurements, to address critical knowledge gaps regarding subsurface contamination distribution, concealed mining-related physical hazards, and hydrological interactions at the ELM site.

The geophysical results revealed a complex subsurface framework comprising a three-tier resistivity and velocity structure, with the ERT surveys identifying pronounced low-resistivity anomalies (1.67 to 12.8 Ωm) beneath previously documented surface contamination hotspots, extending to depths exceeding 19 m and spatially correlating with shaft locations and the slag heap. The integrated magnetic and gravity datasets identified a prominent east-west oriented structural feature, interpreted as a fault-controlled mineralised zone, while the SRT profiles delineated zones of mechanical disturbance indicative of mining-induced subsidence.

Hydrogeochemical analyses of water samples collected from the Edendalspruit stream and the mine discharge point demonstrated that the mine water exhibits elevated electrical conductivity (571 µS/cm), total dissolved solids (396 mg/L), and bicarbonate alkalinity (419 mg/L) relative to upstream conditions, with circumneutral pH (7.38 to 7.68) maintained by carbonate buffering from the local geology. The PHREEQC speciation modelling of downstream water samples revealed undersaturation with respect to calcite (SI = -0.28) and dolomite (SI = 0.38), alongside strong supersaturation of iron oxides and hydroxides (hematite SI = 17.84), indicating that the water chemistry is controlled by carbonate mineral dissolution and iron oxide precipitation.

The salt tracer dilution method quantified the downstream flow rate at 594.2 L/min, while mass balance calculations estimated the mine water discharge contribution at 289.9 L/min, representing a substantial hydrological input. Significantly, despite the elevated concentrations of Pb and Zn in soils (up to 21000 mg/kg and 7300 mg/kg, respectively), these metals remained below detection limits in water samples, suggesting effective immobilisation through adsorption and carbonate precipitation under the prevailing circumneutral, oxidising conditions.

The integrated findings demonstrate that the abandoned mine workings constitute both a persistent source of environmental contamination and a zone of physical instability, with contaminant migration preferentially occurring through structurally controlled pathways, thereby confirming the hypothesis that non-invasive geophysical methods, coupled with hydrochemical and hydrological techniques, can effectively delineate subsurface hazards at

abandoned mine sites. This research provides a comprehensive framework for characterising the interconnected environmental and physical hazards at derelict mine sites, contributing to the limited body of research on physical hazards at abandoned mines and offering a methodological approach applicable to similar sites where historical records are incomplete.